ELEC 221 Tutorial 5: Solutions

DT Fourier series, DT filters, CT Fourier transform

ELEC 221

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1.1 Analysis and synthesis in Python

```python
import numpy as np

def analysis(x, N):
    """c_k=(1/N) sum_n x[n] e^{-jk 2pi n/N}, x = one period"""
    n = np.arange(N); k = n[:, None]
    return (x * np.exp(-2j*np.pi*k*n/N)).sum(axis=1) / N

def synthesis(c, N, n):
    """x[n] = sum_k c_k e^{jk 2pi n/N}, evaluated at integer array n"""
    k = np.arange(N)
    return (c[None, :] * np.exp(2j*np.pi*np.outer(n, k)/N)).sum(axis=1)

# Validate: square wave, N=5, ones at n=-1,0,1
x = np.array([1, 1, 0, 0, 1])                    # n = 0..4
c = analysis(x, 5)
assert np.allclose(c, np.fft.fft(x)/5)           # matches FFT
nn = np.arange(-10, 10)
assert np.allclose(synthesis(c, 5, nn).real, x[nn % 5])  # round trip
```

Check: $c_k = \frac{1}{5}\frac{\sin(3\pi k/5)}{\sin(\pi k/5)}$, $c_0 = \frac{3}{5}$ (the class result).

1.2 Coefficients of $x[n]$ (Figure 1)

From the plot, $x[n]$ has $N = 5$: over one period $x[0] = 1,\ x[\pm 1] = -1,\ x[2] = x[3] = 0$.

Direct analysis over $n = -1, 0, 1$:

$$c_k = \frac{1}{5}\left(1 - e^{jk2\pi/5} - e^{-jk2\pi/5}\right) = \boxed{\frac{1}{5}\left(1 - 2\cos\frac{2\pi k}{5}\right)}$$

Equivalently (by linearity): $x[n] = 2\,\delta_5[n] - s[n]$, with $s[n]$ the 3-impulse square wave and δ_5 the impulse train:

$$c_k = \frac{2}{5} - \frac{1}{5}\frac{\sin(3\pi k/5)}{\sin(\pi k/5)}.$$

- $c_0 = -\frac{1}{5}$ (the mean of x), real and even in k (x is real and even).
- Periodic in k with period 5.

1.3 (a), (b): transformed signals

(a) $x[-n+1]$**.** Time reversal: $c_k \to c_{-k}$ (here $c_{-k} = c_k$). Then shift by $n_0 = 1$: $x[-(n-1)]$.

$$\boxed{b_k = c_{-k}\, e^{-jk2\pi/5} = \tfrac{1}{5}\Big(1 - 2\cos\tfrac{2\pi k}{5}\Big) e^{-jk2\pi/5}}$$

(b) $x_{(3)}[n]$**.** Stretching by $m = 3$ gives period 15; coefficients are scaled by $1/m$ and viewed with period $mN = 15$:

$$\boxed{d_k = \tfrac{1}{3} c_k = \tfrac{1}{15}\Big(1 - 2\cos\tfrac{2\pi k}{5}\Big), \quad k = 0, \ldots, 14}$$

Since c_k has period 5, the pattern repeats three times over $k = 0, \ldots, 14$. The mean drops to $d_0 = -\frac{1}{15}$, because two of every three samples are now zero.

1.3 (c): $x[n] + z[n-2]$, z with $N = 5$

$z[n]$: ones at $n = 0, 1, 2$ (3 impulses centered at $n = 1$). It is the centered square wave $s[n]$ shifted by 1, with $s_k = \frac{1}{5}\frac{\sin(3\pi k/5)}{\sin(\pi k/5)}$.

$z[n-2]$ is $s[n]$ shifted by $1 + 2 = 3$. Both signals have period 5, so we add coefficients directly:

$$\boxed{e_k = c_k + s_k\, e^{-jk6\pi/5} = \tfrac{1}{5}\left(1 - 2\cos\tfrac{2\pi k}{5}\right) + \tfrac{1}{5}\frac{\sin(3\pi k/5)}{\sin(\pi k/5)}\, e^{-jk6\pi/5}}$$

with $e_0 = -\frac{1}{5} + \frac{3}{5} = \frac{2}{5}$.

Check (one period, $n = 0..4$): $x = (1, -1, 0, 0, -1)$, $z[n-2] = (0, 0, 1, 1, 1)$, sum $= (1, -1, 1, 1, 0)$, mean $= \frac{2}{5}$ ✓

1.3 (d): same, but $z[n]$ has $N = 6$

Now x has period 5 and z has period 6: the sum has period $\mathrm{lcm}(5, 6) = 30$. Express both with $N = 30$ (harmonics $k = 0, \ldots, 29$).

$x[n]$ (period 5 $\Rightarrow$ stretched by 6 in frequency): $a_k = \begin{cases} c_{k/6}, & 6 \mid k \\ 0, & \text{else} \end{cases}$

$z[n-2]$ (period 6): ones at $n = 2, 3, 4$, centered at 3. Over its own period $N = 6$:

$$f_m = \tfrac{1}{6} \frac{\sin(\pi m/2)}{\sin(\pi m/6)} e^{-jm2\pi \cdot 3/6} = \tfrac{1}{6} \frac{\sin(\pi m/2)}{\sin(\pi m/6)} (-1)^m, \quad f_0 = \tfrac{1}{2}$$

On the $N = 30$ grid: $g_k = \begin{cases} f_{k/5}, & 5 \mid k \\ 0, & \text{else} \end{cases}$

$$\boxed{h_k = a_k + g_k, \quad k = 0, \ldots, 29}$$

Non-zero only at multiples of 5 and 6. The two sets overlap at $k = 0$: $h_0 = -\frac{1}{5} + \frac{1}{2} = \frac{3}{10}$ (the mean).

2.1 Output of the filter in Figure 2

From the figure, the passband is

$$\frac{\pi}{5} \le |\omega| \le \frac{3\pi}{5} \pmod{2\pi}, \qquad H(e^{j\omega}) = 1 \text{ there, } 0 \text{ elsewhere.}$$

Each component of $x[n] = 2 + \sin(\frac{7\pi}{8}n + \frac{\pi}{5}) + \sin(\frac{\pi}{8}n)$ is scaled by H at its frequency:

Component	ω	$\lvert H\rvert$
2 (DC)	0	0
$\sin(\frac{7\pi}{8}n + \frac{\pi}{5})$	$0.875\pi > 0.6\pi$	0
$\sin(\frac{\pi}{8}n)$	$0.125\pi < 0.2\pi$	0

$$\boxed{y[n] = 0}$$

No component lies in the passband. (The filter is a band-pass.)

2.2 IIR impulse response: $y[n] - 0.5\,y[n-1] = x[n]$

Rewrite as $y[n] = 0.5\,y[n-1] + \delta[n]$, with $y[-1] = 0$ and $x = \delta$:

n	0	1	2	3	4
$h[n]$	1	0.5	0.25	0.125	0.0625

Pattern: $h[n] = (0.5)^n\,u[n]$ — infinite duration, hence IIR.

Stability:

$$\sum_n |h[n]| = \sum_{n=0}^{\infty} 0.5^n = \frac{1}{1-0.5} = 2 < \infty \;\Rightarrow\; \text{BIBO stable.}$$

(In general $h[n] = a^n u[n]$ is stable iff $|a| < 1$.) Frequency response: $H(e^{j\omega}) = \dfrac{1}{1 - 0.5\,e^{-j\omega}}$.

3.1 Spectrum of the piecewise signal

$x(t) = -1$ on $[-1, 1)$, 2 on $[1, 2)$:

$$\begin{aligned} X(j\omega) &= -\int_{-1}^{1} e^{-j\omega t}\,dt + 2\int_{1}^{2} e^{-j\omega t}\,dt \\ &= -\frac{e^{j\omega} - e^{-j\omega}}{j\omega} + \frac{2\left(e^{-j\omega} - e^{-2j\omega}\right)}{j\omega} \end{aligned}$$

$$\boxed{X(j\omega) = \frac{-e^{j\omega} + 3e^{-j\omega} - 2e^{-2j\omega}}{j\omega}}$$

Check: numerator $\to -1 + 3 - 2 = 0$ as $\omega \to 0$, so $X(0) = \int x\,dt = -2 + 2 = 0$ ✓

3.2 $x(t) = e^{-t}$ on $[-1, 1)$

$$X(j\omega) = \int_{-1}^{1} e^{-(1+j\omega)t} dt = \frac{e^{1+j\omega} - e^{-(1+j\omega)}}{1+j\omega} = \boxed{\frac{2\sinh(1+j\omega)}{1+j\omega}}$$

Periodic extension ($T = 2$, $\omega_0 = \pi$):

$$c_k = \frac{1}{2}\int_{-1}^{1} e^{-t} e^{-jk\pi t} dt = \frac{1}{T} X(jk\omega_0) = \frac{(-1)^k \sinh 1}{1+jk\pi}$$

using $\sinh(1 + jk\pi) = (-1)^k \sinh 1$.

Takeaway: $c_k = \frac{1}{T} X(jk\omega_0)$. The Fourier series coefficients are *samples* of the spectrum of one period, spaced ω_0 apart and scaled by $1/T$.

3.3 Spectrum of periodic signals

(a) $\int \cos t\, e^{-j\omega t} dt = \frac{1}{2}\int \left(e^{j(1-\omega)t} + e^{-j(1+\omega)t}\right) dt$

This does not converge: $\cos t$ is not absolutely integrable, and the integrand never decays. We get stuck evaluating $\lim_{T\to\infty}$ of sin terms.

(b) $\cos t = \frac{1}{2}(e^{jt} + e^{-jt})$ contains only $\omega = \pm 1$, so we expect energy *only* at $\omega = \pm 1$.

(c) Try $X(j\omega) = 2\pi\delta(\omega - \omega_0)$ in the synthesis equation:

$$\frac{1}{2\pi}\int 2\pi\delta(\omega-\omega_0)e^{j\omega t}d\omega = e^{j\omega_0 t} \;\Rightarrow\; e^{j\omega_0 t} \overset{\mathcal{F}}{\longleftrightarrow} 2\pi\delta(\omega-\omega_0)$$

By linearity: $\boxed{\cos t \;\longleftrightarrow\; \pi\left[\delta(\omega-1)+\delta(\omega+1)\right]}$

(d) For $x(t) = \sum_k c_k e^{jk\omega_0 t}$:

$$\boxed{X(j\omega) = 2\pi \sum_{k=-\infty}^{\infty} c_k\, \delta(\omega - k\omega_0)}$$

An impulse train in frequency, with areas $2\pi c_k$ at the harmonics.